Using Machine Learning to Identify Effective Municipal Education Management Practices in Brazil

Gabriel Couto

Stanford University

CS129: Applied Machine Learning

Andrew Ng & Younes Mourri

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Introduction

As discussed by Dalmon (2025), large-scale interventions to improve learning outcomes are challenging due to the complexity of working with multiple policymakers and adapting to diverse realities across, cities, and regions. Middle-tier governments, such as districts and municipalities, play an important role in addressing local needs, yet research on the influence of local education departments on student achievement in developing countries remains limited.

In 2025, the PhD thesis of Danilo Leite Dalmon (Stanford GSE), who suggested this study and provided initial advice, began exploring the correlation between practices of local education departments and primary school achievement in Brazil, which was chosen due to its large number of municipalities (over 5,500) that vary significantly in socioeconomic status, demographics, and geography (p. 1–2). Building on this work, we examine which practices of municipal education secretaries in Brazil influence student math performance in public schools, measured by SAEB 2023 scores. We use a Random Forest regression model to identify key predictors through feature importance, correlation analysis, and sequential feature selection.

Related work

Cilliers et al. (2022) serve as the main methodological reference for this project. The authors collected data from District Education Officers across 184 Local Government Authorities in Tanzania, including management practices such as decision making, performance review, and interactions with schools, and combined it with PSLE scores, socioeconomic, and census data. They applied a Random Forest model, averaging multiple trees built on random subsamples, to identify which managerial practices best predict academic performance. Variable importance was assessed to measure each predictor’s contribution to model performance.

We also draw on Tertulino & Almeida (2025) as a reference. Based on SAEB data, their study identifies predictors of student performance, focusing on socioeconomic factors. In contrast, our study centers on municipal governance practices. They also compared multiple machine learning models, including XGBoost and CatBoost, and found that Random Forest produces the most accurate predictions (p. 1).

Dataset

This project uses data from the Basic Education Assessment System (SAEB), which includes questionnaire items on teacher training, interactions with school principals, and other administrative practices reported by municipal Secretaries of Education across Brazil. The questionnaire responses were converted into numerical values (e.g., yes/no coded as 0/1).

The original dataset included 5,568 municipalities and 176 variables. Data cleaning involved removing open-ended and non-relevant variables (e.g., administrator gender or race), as these were not considered predictive of student performance. We also excluded municipalities with missing questionnaires or substantial missing data.

The final dataset consists of 1,267 municipalities, 102 independent variables, and the SAEB 2023 math score as the dependent variable, measured as a weighted average of school-level scores derived from individual student performance.

The data was split into training (60%), validation, and test (20%) sets using random sampling with a fixed seed to ensure reproducibility. The validation set was used for hyperparameter tuning, whereas the test set was reserved for the final model evaluation.

Methods & discussion

Random Forest was the main method used in this study based on Tertulino & Almeida's (2025) recommendation. This algorithm averages the predictions of multiple decision trees, allowing it to capture complex, non-linear relationships without strong parametric assumptions. It is well suited to settings with many predictors and noisy survey-based data, as it reduces variance and handles irrelevant variables effectively.

$$\hat{f}(x) = \frac{1}{B} \sum_{b=1}^{B} T_b(x)$$

$\hat{f}(x)$: predicted outcome (SAEB math score)
B: number of trees (n_estimators)
$T_b(x)$: prediction from tree b
x: vector of predictors (municipal practices)

The first Random Forest model was trained using all initial variables to establish a baseline performance and compute feature importance, which measures how much each variable contributes to the model's predictions.

These results were used to identify the most relevant variables and guide subsequent feature selection and model refinement. To choose the best parameters, we used *for loops* to train models with different maximum depths (Figure 1) and numbers of trees (Figure 2), selecting those that maximized R^2. After training the model with the optimal parameters, the R^2 was around 0.87 on the training set and around 0.14 on the test set.

Figure 1

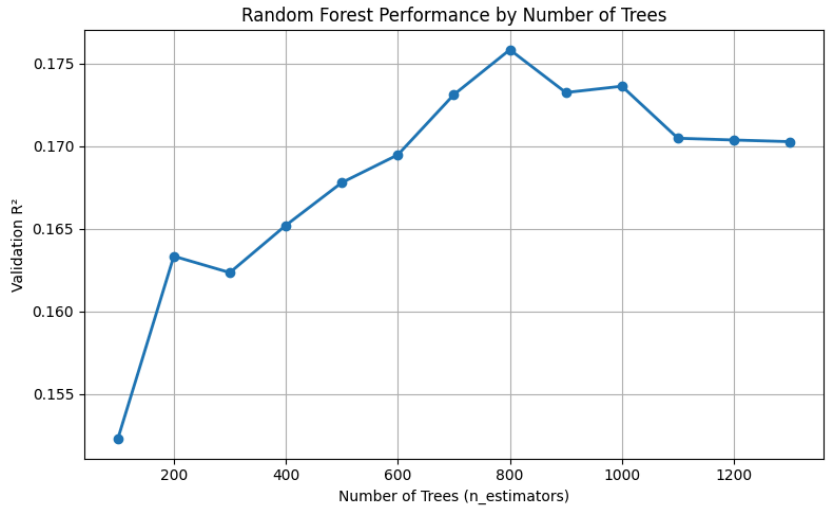
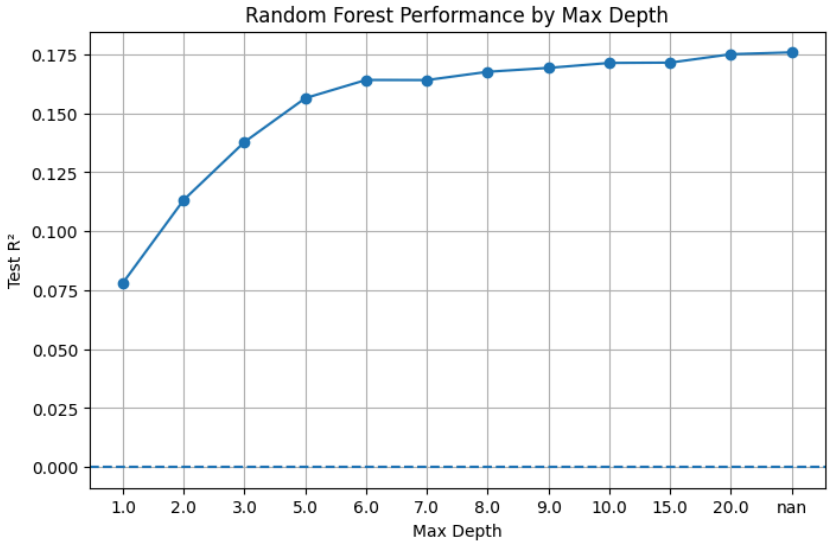


Figure 2



The residual analysis shows a large and random dispersion around zero, indicating substantial prediction error on unseen data (Figure 3). Combined with the large gap between training and test R^2, this suggests that the model overfits the training data, capturing noise rather than generalizable patterns, likely due to the high number of variables. This is further confirmed by the plot of actual versus predicted values (Figure 4).

Figure 3

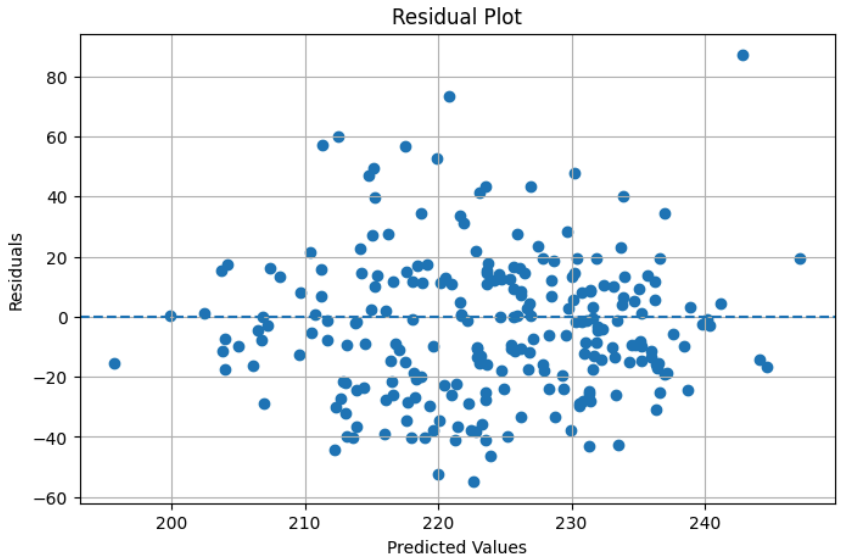
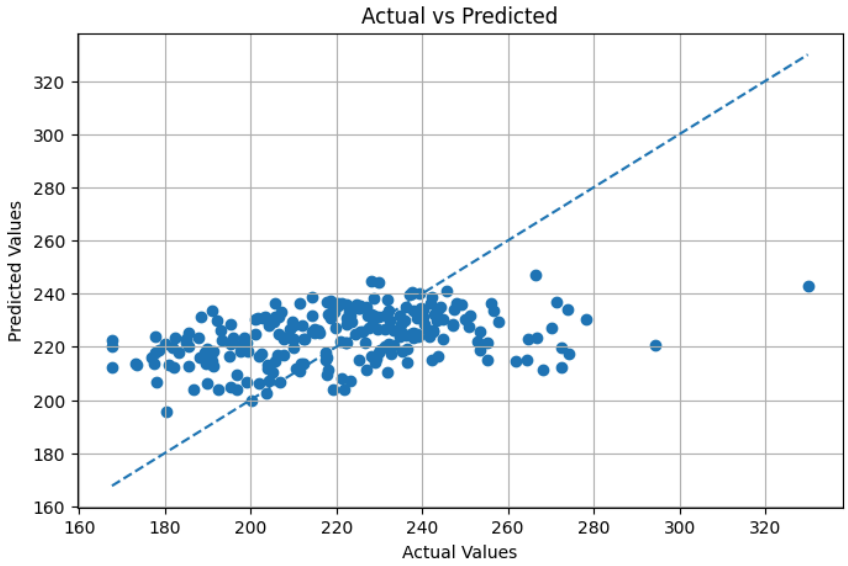
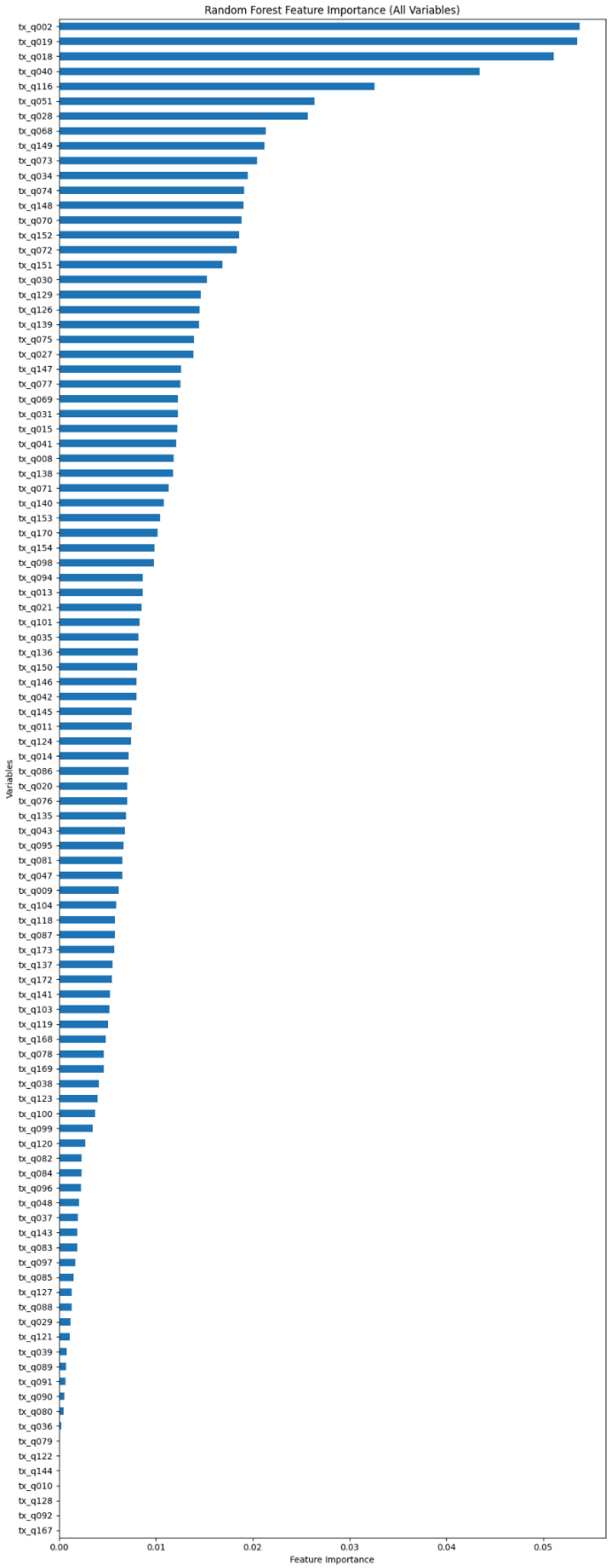


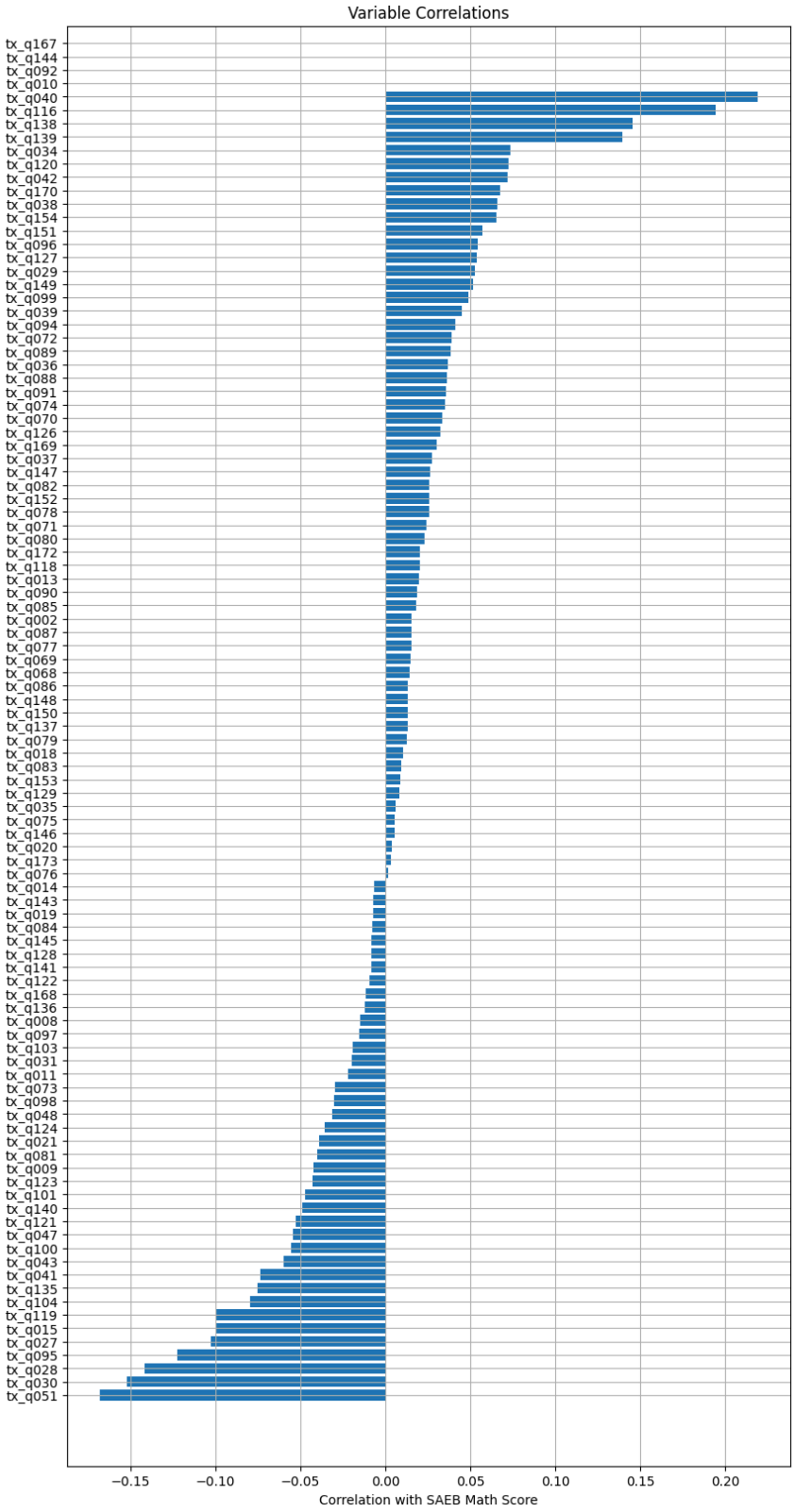
Figure 4



After the baseline model, we calculated feature importance (Figure 5) and correlation (Figure 6), and filtered the top variables based on minimum thresholds (importance ≥ 0.005 and correlation ≥ 0.03). Using the remaining variables, we applied Sequential Feature Selection on the top 20 candidates (due to computational constraints). This method iteratively builds regression models by adding variables and evaluating their performance. At each step, the combination of variables that maximizes predictive accuracy (R^2) is retained. This process identifies the subset of features that provides the best-performing model while reducing dimensionality.

Figure 5


Random Forest Feature Importance (All Variables)
Variables
tx_q002
tx_q019
tx_q018
tx_q040
tx_q116
tx_q051
tx_q028
tx_q068
tx_q149
tx_q073
tx_q034
tx_q074
tx_q148
tx_q070
tx_q152
tx_q072
tx_q151
tx_q030
tx_q129
tx_q126
tx_q139
tx_q075
tx_q027
tx_q147
tx_q077
tx_q069
tx_q031
tx_q015
tx_q041
tx_q008
tx_q138
tx_q071
tx_q140
tx_q153
tx_q170
tx_q154
tx_q098
tx_q094
tx_q013
tx_q021
tx_q101
tx_q035
tx_q136
tx_q150
tx_q146
tx_q042
tx_q145
tx_q011
tx_q124
tx_q014
tx_q086
tx_q020
tx_q076
tx_q135
tx_q043
tx_q095
tx_q081
tx_q047
tx_q009
tx_q104
tx_q118
tx_q087
tx_q173
tx_q137
tx_q172
tx_q141
tx_q103
tx_q119
tx_q168
tx_q078
tx_q169
tx_q038
tx_q123
tx_q100
tx_q099
tx_q120
tx_q082
tx_q084
tx_q096
tx_q048
tx_q037
tx_q143
tx_q083
tx_q097
tx_q085
tx_q127
tx_q088
tx_q029
tx_q121
tx_q039
tx_q089
tx_q091
tx_q090
tx_q080
tx_q036
tx_q079
tx_q122
tx_q144
tx_q010
tx_q128
tx_q092
tx_q167
0.00
0.01
0.02
0.03
0.04
0.05
Feature Importance


Variable Correlations
tx_q167
tx_q144
tx_q092
tx_q010
tx_q040
tx_q116
tx_q138
tx_q139
tx_q034
tx_q120
tx_q042
tx_q170
tx_q038
tx_q154
tx_q151
tx_q096
tx_q127
tx_q029
tx_q149
tx_q099
tx_q039
tx_q094
tx_q072
tx_q089
tx_q036
tx_q088
tx_q091
tx_q074
tx_q070
tx_q126
tx_q169
tx_q037
tx_q147
tx_q082
tx_q152
tx_q078
tx_q071
tx_q080
tx_q172
tx_q118
tx_q013
tx_q090
tx_q085
tx_q002
tx_q087
tx_q077
tx_q069
tx_q068
tx_q086
tx_q148
tx_q150
tx_q137
tx_q079
tx_q018
tx_q083
tx_q153
tx_q129
tx_q035
tx_q075
tx_q146
tx_q020
tx_q173
tx_q076
tx_q014
tx_q143
tx_q019
tx_q084
tx_q145
tx_q128
tx_q141
tx_q122
tx_q168
tx_q136
tx_q008
tx_q097
tx_q103
tx_q031
tx_q011
tx_q073
tx_q098
tx_q048
tx_q124
tx_q021
tx_q081
tx_q009
tx_q123
tx_q101
tx_q140
tx_q121
tx_q047
tx_q100
tx_q043
tx_q041
tx_q135
tx_q104
tx_q119
tx_q015
tx_q027
tx_q095
tx_q028
tx_q030
tx_q051
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
Correlation with SAEB Math Score

Similarly to the first Random Forest model, we performed hyperparameter tuning to select the maximum depth (Figure 7) and number of trees (Figure 8). The final model, using the selected variables and parameters, achieved an R^2 of approximately 0.16 on the training set and 0.08 on the test set (Figure 9). Although this is lower than the baseline model, the smaller gap between training and test performance suggests reduced variance and less overfitting. This is supported by the residual analysis, which shows residuals centered around zero with no strong systematic pattern, indicating limited bias. However, the wide dispersion and slight clustering of residuals suggest substantial prediction error and limited predictive precision (Figure 10).

As a baseline, we estimated a linear regression model using the full set of input variables (i.e., all original predictors used in the initial Random Forest model). The model achieved an R^2 of approximately 0.04 on the test set. While the Random Forest model provides better predictive performance (test $R^2 \approx 0.08$), its overall predictive power remains limited.

Figure 7

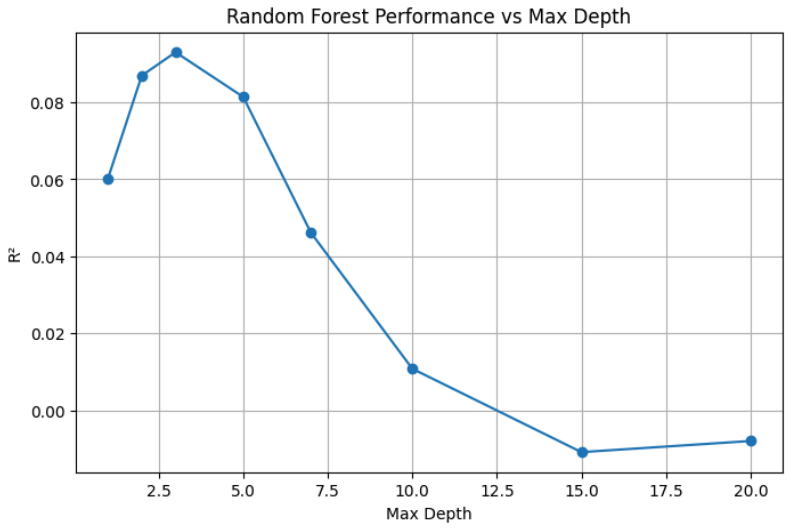


Figure 8

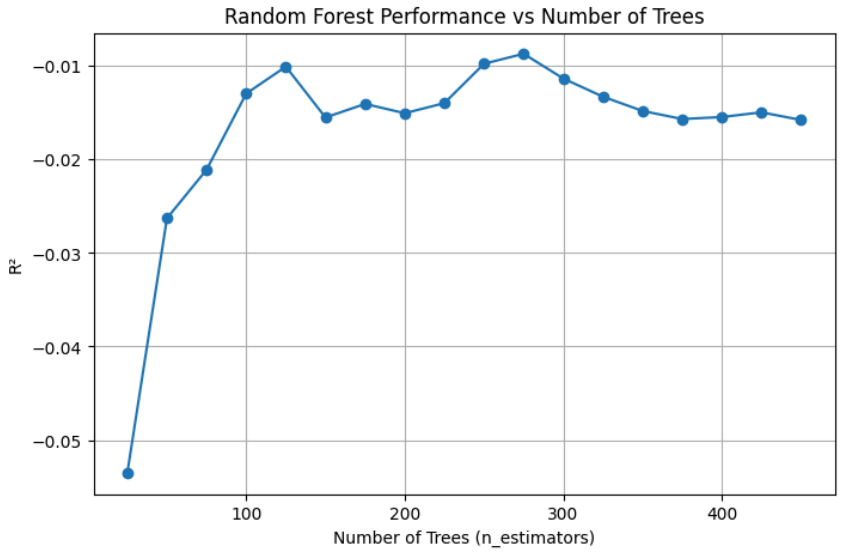


Figure 9

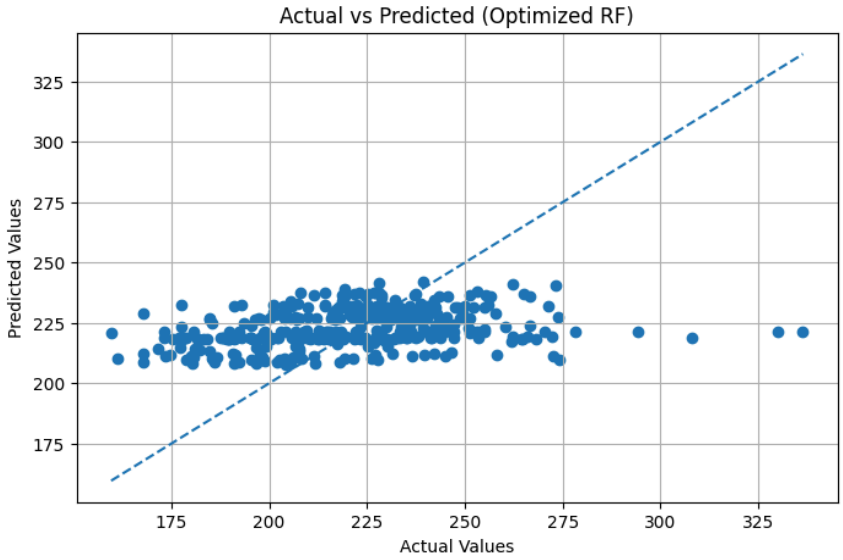
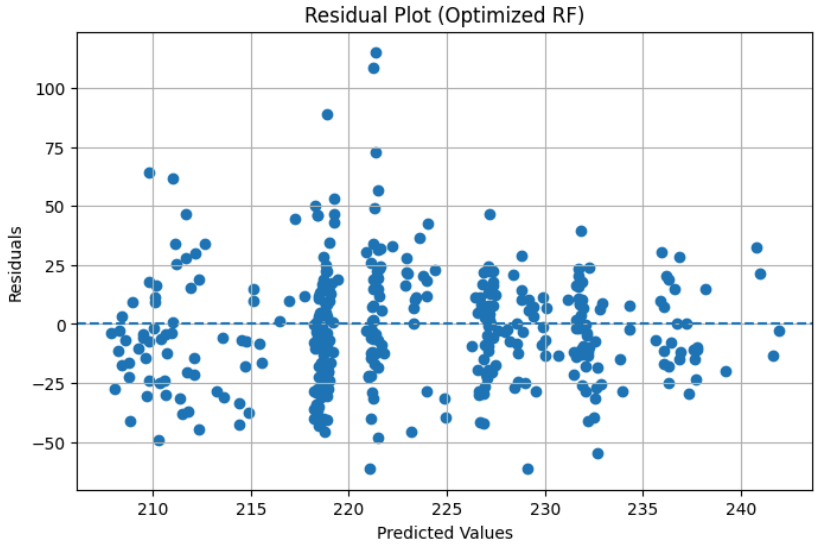


Figure 10



Conclusion & future work

After estimating the second Random Forest model, we identified the main predictors of student performance as the municipal education secretary's years of experience, the use of standardized curriculum developed by external organizations (NGO's, companies or other institutions), weather if the secretary is charge of deciding how much, when, and how use financial resources available in the annual education budget, the existence of a municipal education system responsible for defining municipal norms and practices as well as overseeing public programs and an education forum, as well as the following practices: collaboration with special education institutions, principal selection based on management experience, budget allocation priority given to school construction and material purchase, the use of SAEB scores to guide school rewards.

The relatively low R^2 is expected given the complexity of educational outcomes and the limited scope of survey-based administrative variables, which capture only part of the factors influencing student performance. Future research could explore alternative model specifications

and more systematic hyperparameter tuning to improve predictive performance. Additionally, validating the model across different years or datasets would help assess its robustness and generalizability.

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Extended Abstract

Tertulino, R., & Almeida, R. (2025). *A Multi-level Analysis of Factors Associated with Student Performance: A Machine Learning Approach to the SAEB Microdata.* https://doi.org/10.48550/arXiv.2510.22266